

Problem Statement vs Objectives in Data Analytics Projects: A Hospital Case Study

In a data project, the **Problem Statement** and the **Objectives** play critical roles in defining the *scope* and *purpose* for analysis. This document outlines their differences and includes examples of good vs. bad problem statements using a hospital scenario.

1. What Is a Problem Statement?

In its simplest form, a **problem statement** defines the pain point you hope to solve or the impact you hope to make with your work or the opportunities that the project seeks to address. A problem statement is articulated at the **beginning of a project** to set the stage for analysis.

A problem statement provides **context** and explains **why the project exists**.

Characteristics of a Good Problem Statement:

- Specific and descriptive.
- Clear and concise.
- Focuses on the core issue affecting the organization.
- Sets the foundation for analysis without diving into specific questions.

What needs to be Mentioned in a Problem Statement:

- The *specific problem*
- A *numerical value* to represent the magnitude of the problem
- Location (*if available*)
- Timeframe (*if available*)
- *Impact* of the problem (Why is it a problem? Were there any bad consequences because of it?)

Bad vs. Good Problem Statements

Example 1: Increasing ER Wait Times

Bad Problem Statement:

"ER wait times are long."

Why it's bad:

- It's too vague, lacks context, and doesn't describe the scope or impact of the issue.

Good Problem Statement:

"Hospital X has reported a **25% increase** in emergency room (ER) wait times over the **last year**, leading to a **rise in patient complaints** and a **decline in satisfaction scores**."

Why it's good:

- Clearly defines the issue using a numeric value (*25% increase in ER wait times*).
- Provides context, including location and timeframe (*Hospital X, last year,...*).
- Lists the impact of the issue (*rise in complaints and satisfaction decline*).

Example 2: Declining Patient Satisfaction

Bad Problem Statement:

"Patient satisfaction scores are low."

Why it's bad:

- It doesn't specify how low, over what period, or what might be causing the decline.

Good Problem Statement:

"Over the **past six months**, Hospital Y's **patient satisfaction scores** have **dropped**"

by **15%**, with survey feedback indicating dissatisfaction with **long wait times for specialist appointments**. This decline in satisfaction is contributing to an **increase in patient complaints** and a **potential loss of returning patients**."

Why it's good:

- Clearly defines and quantifies the issue (*15% drop in satisfaction*). • Provides context (*Hospital Y, past six months, dissatisfaction with specialist wait times, ...*).
- Lists the impact of the issue (*increase in patient complaints, loss of returning patients*).

2. What Are Objectives/Approaches?

Objectives break the problem into specific, actionable queries that guide the data analysis. They focus on **what needs to be analyzed and measured** to solve the problem.

Characteristics of Good Objectives:

- Specific, measurable, and actionable.
- Directly address aspects of the problem statement.
- Help identify insights and guide decision-making.

Bad vs. Good Objectives

Example 1: ER Wait Times

Bad Objective:

"Why are wait times bad?"

Why it's bad:

- It's vague and subjective.
- It doesn't focus on any specific metrics or variables.

Good Objective:

"What is the **average ER wait time** over the **last 12 months**, and how does it vary **by day of the week**?"

Why it's good:

- Focuses on measurable metrics (*average wait time*).
- Includes specific dimensions for analysis (*day of the week*).

Example 2: Patient Satisfaction

Bad Objective:

"How can we make patients happier?"

Why it's bad:

- It's too broad and subjective, with no clear direction for analysis.

Good Objective:

"What **percentage of specialist appointments** had **wait times exceeding 30 minutes** in the **past six months**, and how do these wait times **correlate with patient satisfaction scores**?"

Why it's good:

- Focuses on measurable metrics (*percentage of long waits*).
- Explores relationships (*correlation with satisfaction scores*).

3. Key Takeaways

- A **problem statement** sets the high-level direction for a data analytics project by explaining the **why** behind the analysis.
- **Objectives** focus on specific, actionable queries that guide the **how** of the analysis.
- A **bad problem statement or objective** is vague, broad, and lacks

focus, while a **good one** is clear, measurable, and actionable.

Tips for Writing Problem Statements and Objectives

1. Keep the Problem Statement Specific:

- Avoid diving into analysis specifics.
- Set the scene and define the overall challenge.

2. Make Objectives SMART:

- **Specific:** Focus on a *single issue* or metric.
- **Measurable:** Use *quantifiable metrics*.
- **Attainable:** Ensure the question can be answered with *available data*.
- **Reproducible:** Define *clear terms* for consistency.
- **Time-bound:** Specify a *timeframe*.

Partner Practice Exercise

In this exercise, you will simulate a requirement gathering session to address a business problem. One participant will act as the business stakeholder while the other will take on the role of the analyst. Your goal is to fully understand the problem, ask pertinent questions, and craft a well-defined problem statement given a generic business scenario.

Scenario: Streamo, a streaming service provider, has been experiencing an increasing rate of customer churn.